

Bluetooth Channel Sounding

Mode-3 Combined Round-Trip Time and Phase-Based Ranging

The mirrored sub-slot order, same-step coherence of the two observables, joint estimation and calibration, per-channel ambiguity resolution, same-step integrity, and the air-time economics that decide when to use it

This report traces the complete measurement chain of a Bluetooth Channel Sounding Mode-3 step: the extended packet in which a CS_SYNC exchange and a tone exchange share a single step, the mirrored ordering of the two sub-slots in the initiator and reflector bursts, and the two observables — a round-trip time and a set of phase correction terms — that the step produces together. It derives the step timing and shows the central structural result: the mirrored order leaves the phase measurement's reciprocal interval *identical* to Mode-2's while lengthening the reflector turnaround that governs the round-trip-time clock-scale error, so Mode-3 buys a co-located round-trip time at a cost paid entirely by the coarse observable that can most easily absorb it. It develops what same-step acquisition makes newly possible — per-channel unwrap seeding that is unconditionally safe across the whole band, joint calibration of loop delay and loop phase on one hardware path, a joint delay estimator over the combined packet and tone aperture, and a per-channel integrity check that binds the predictable tone to the unpredictable packet in the same step. A worked example consistent with the campaign report's configuration quantifies both chains, and the report closes with an error budget, an air-time trade study, a verification plan, and the boundary between specified behaviour and implementation choice.

Primary references	Core 6.3, Vol. 6, Part H, Sec. 4.3.4; Vol. 1, Part A, Secs. 9.1–9.2
Air-interface timing	Core 6.3, Vol. 6, Part H, Secs. 4.3 and 4.5
Host interface	Core 6.3, Vol. 4, Part E, LE CS Subevent Result event
Companion reports	<i>Mode-0 Frequency Estimation</i> , Rev. 4.0; <i>Mode-1 RTT Estimation</i> , Rev. 1.0; <i>Mode-2 Phase-Based Ranging</i> , Rev. 1.0; <i>Campaign Configuration and Scheduling</i> , Rev. 1.0
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Abstract

Mode-3 is the combined step mode of Bluetooth Low Energy Channel Sounding [Core 6.3, Vol. 6, Part H, Sec. 4.3.4]. Within one step on one channel, the initiator transmits an extended packet consisting of a CS_SYNC packet, a guard interval, and a tone sequence covering every antenna path; after the interlude period the reflector transmits the same content in the *reverse* order, tone sequence first and CS_SYNC last. The step therefore yields both a Mode-1-like round-trip time and a Mode-2-like set of phase correction terms, acquired on the same channel, through the same hardware path, within a few hundred microseconds of each other.

The estimation theory of the two observables is not new: it is developed in the companion Mode-1 and Mode-2 reports and applies here unchanged. What is new in Mode-3 is everything that follows from the two observables being *co-located in one step*, and that is the subject of this report. Four consequences are developed quantitatively. The mirrored sub-slot order is shown to be a deliberate optimisation: the reciprocal interval between the two tone measurements is $T_{TONE} + T_{RD} + T_{IP2}$, exactly the Mode-2 value, so adding the packets costs the phase chain nothing, while the reflector turnaround grows to $T_{SY} + 2T_{GD} + 2T_{TONE} + T_{RD} + T_{IP2}$ and makes the round-trip-time chain roughly two and a half times more sensitive to residual clock-scale error than Mode-1. A round-trip time available on the *same* channel as each phase sample converts unwrapping from a fragile sequential accumulation into a set of independent nearest-integer decisions that remain safe across any gap the 2.4 GHz band can present. The loop group delay of the timing chain and the loop phase response of the phase chain are properties of one hardware path traversed once, so they can be calibrated jointly and checked against each other. And because the unpredictable CS_SYNC and the fully predictable tone occupy the same step, the packet's integrity metric bounds an attack on the tone in a way that interleaved Mode-1 and Mode-2 steps cannot.

A worked example at 2440 MHz, using the configuration of the companion campaign report, gives a step duration of $274 \mu s$, a phase-chain uncertainty of 1.57 cm and a timing-chain uncertainty of 12.8 cm, and yields a 3σ cross-observable consistency gate of 39 cm. An air-time study shows Mode-3 to be 23.7% cheaper in occupancy than a Mode-1 and a Mode-2 step taken separately, but 16.6% more expensive than the sparse-sub-mode campaign pattern it would replace.

Status and interpretation. Normative claims are tied to the cited Bluetooth Core Specification sections. Estimators, joint-estimation formulations, calibration schemes, quality metrics and aggregation rules are representative implementation methods. Timing parameter value lists and report field encodings should be read from the adopted specification revision in force; where this report states them, they are given to make the equations concrete, and section numbering within Volume 6, Part H has changed between Core revisions. Two internal inconsistencies in the existing report series that this document had to resolve are flagged explicitly in Section 3. The Bluetooth specification and adopted test specifications remain authoritative for conformance.

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1 Scope and measurement objective

Bluetooth Channel Sounding uses four step modes, defined in Core 6.3, Volume 6, Part H, Sections 4.3.1–4.3.4. Mode-0 establishes the relative frequency and timing scale between the two devices. Mode-1 uses that scale to measure a round-trip time. Mode-2 measures phase on tone exchanges. Mode-3 performs both measurements within a single step.

The Mode-3 measurement objective is the union of the Mode-1 and Mode-2 objectives, with one addition that is easy to overlook and is the reason the mode exists: the two observables carry a *binding* to each other. Per step, the reported set is

- the initiator’s and reflector’s timing differences, from which a round-trip time is formed exactly as in Mode-1 [Core 6.3, Vol. 4, Part E, LE CS Subevent Result];
- the per-antenna-path phase correction terms and tone quality indicators from both devices, from which a two-way phase is formed exactly as in Mode-2; and
- the fact, guaranteed by the step structure rather than by inference, that both were measured on the same channel, through the same antenna path and gain state, within one step.

1.1 What this report does and does not repeat

The estimation theory of each observable in isolation is developed in the companion reports and is not reproduced here. Time-of-arrival estimation from a CS_SYNC packet — matched filtering, sub-sample interpolation, maximum likelihood, group-delay regression, first-path detection under multipath, and the Cramér–Rao bound relating timing variance to root-mean-square bandwidth and accumulated signal-to-noise ratio — is the subject of the Mode-1 report. Per-slot phase estimation and the conversion of a set of per-channel phase correction terms into a distance — coherent averaging, frequency estimation, front-end impairments, slope regression, inverse-transform channel impulse response formation, super-resolution and ambiguity — is the subject of the Mode-2 report. Both apply to Mode-3 without modification.

This report covers what is specific to Mode-3: the step structure and its timing consequences (Sections 5–6), what same-step acquisition makes newly possible (Sections 7–8), the combined error budget and the cross-observable consistency gate (Sections 9 and 12), same-step integrity (Section 10), and the air-time economics that decide whether to configure Mode-3 at all (Section 11).

Three properties that are frequently confused. (1) Mode-3 is *not* simply a Mode-1 step and a Mode-2 step concatenated: the sub-slot order is mirrored between the two devices, and that mirroring is what keeps the phase chain’s reciprocal interval as short as Mode-2’s. (2) Mode-3 does not improve either observable’s precision — a Mode-3 round-trip time is no better than a Mode-1 round-trip time on the same configuration, and its phase samples are no better than Mode-2’s. What improves is everything that depends on the two being *jointly* available. (3) Mode-3 is not free: its step is longer than either constituent, and whether it is cheaper than the alternative depends entirely on how often the campaign would otherwise schedule a round-trip-time step.

2 Normative framework and terminology

Table 1: Normative map for the Mode-3 combined measurement. Section numbering should be confirmed against the specification revision in force.

Source	What it defines	Question answered
Core 6.3, Vol. 1, Part A, Sec. 9.1	CS procedure architecture; ordering of calibration and ranging steps	Why must Mode-0 precede Mode-3?
Core 6.3, Vol. 1, Part A, Sec. 9.2	Distance estimation from phase and amplitude information	What is the intended use of the phase observable?
Core 6.3, Vol. 6, Part H, Sec. 4.3.4	Mode-3 step structure: the extended packet and its sub-slot order in each direction	What is transmitted, in which order, and when?
Core 6.3, Vol. 6, Part H, Sec. 4.3.2	Mode-1 step, whose packet exchange Mode-3 embeds	How is the round-trip time formed?
Core 6.3, Vol. 6, Part H, Sec. 4.3.3	Mode-2 step, whose tone exchange Mode-3 embeds	How is the phase observable formed?
Core 6.3, Vol. 6, Part H, Sec. 4.3	T_{SY} , T_{GD} , T_{SW} , T_{PM} , T_{RD} , interlude and T_{FCS} value sets	Which step timings may be configured?
Core 6.3, Vol. 6, Part H, Sec. 4.4	CS_SYNC construction: access address, trailer, sounding and random sequences	What waveform is the timing receiver matched to?
Core 6.3, Vol. 6, Part H, Sec. 4.5	Antenna path handling, permutation, tone extension slot	Which physical path does a tone slot correspond to?
Core 6.3, Vol. 6, Part A, Sec. 3.5	Frequency measurement and generation in CS	On what carrier are the packet and tone expected?
Core 6.3, Vol. 4, Part E, LE CS Subevent Result event	Per-step timing, PCT, TQI and NADM reporting	How is the measurement exposed to the host?
RFPHY.TS / LL.TS, CS step Mode-3 clauses	Instrument-level verification of the combined step	How is the behaviour tested?

Table 2: Principal symbols. Symbols shared with the companion reports keep their meaning.

Symbol	Meaning	Units
T_{SY}	CS_SYNC packet duration	s
T_{GD}	Guard interval between CS_SYNC and tone sequence	s
T_{TONE}	Tone-sequence duration, $(T_{SW} + T_{PM})(N_{AP} + 1)$	s
T_{SW}, T_{PM}	Antenna path switch and phase measurement periods	s
T_{RD}	Transmitter ramp-down period	s
T_{IP2}	Interlude period for Mode-3 [Core 6.3, Vol. 6, Part H, Sec. 4.3.4]	s
N_{AP}	Number of antenna paths	—
ΔT	Interval between the reciprocal tone measurements of one path	s
B_R	Reflector turnaround interval, CS_SYNC arrival to departure	s
A_I	Initiator-measured round interval $ToA_I - ToD_I$	s
RTT, ToF	Round-trip time and one-way time of flight	s
$\Theta(f_k)$	Two-way phase sum on channel k	rad
$H[k]$	Assembled frequency-domain channel estimate	—
D_{loop}	Total calibrated loop group delay of the four chains	s
$\Phi_{loop}(f)$	Total calibrated loop phase response	rad
ϵ	Residual fractional frequency offset after Mode-0	—
$\hat{d}_{RTT}, \hat{d}_{\Theta}$	Range from the timing and phase chains	m
TQI, NADM	Tone quality indicator; Normalized Attack Detector Metric	index

3 Two inconsistencies in the series, and how this report resolves them

Two points had to be settled before the timing algebra of Section 5 could be written, because the existing material in this series disagrees with itself. Both are recorded here rather than buried, because both affect numbers that other documents depend on.

(1) The interlude is T_{IP2} — resolved. This was an open question when this report was first drafted: the Mode-3 timing artwork labelled the interlude $T_{IP,2}$ while the companion campaign report used T_{IP1} . Core 6.3 settles it. Table 4.10 of the Channel Sounding part gives the Mode-3 step as $T_{SY} + T_{GD} + (T_{SW} + T_{PM})(N_{AP} + 1)$, T_{RD} , **IP2**, $(T_{SW} + T_{PM})(N_{AP} + 1) + T_{GD} + T_{SY}$, T_{RD} [Core 6.3, Vol. 6, Part H, Sec. 4.3.4], and the accompanying text states that “the duration of T_{IP2} is defined in Section 4.3.3”. The artwork is correct and the campaign report is wrong. Since T_{IP1} and T_{IP2} share the value set $\{10, 20, 30, 40, 50, 60, 80, 145\} \mu s$, no number in this report changes; only the parameter an implementation must read does. This report writes the interlude as T_{IP2} throughout.

(2) The guard intervals are missing from the campaign report’s step duration. The campaign report gives

$$T_{step,3} = 2T_{SY} + 2(T_{SW} + T_{PM})(N_{AP} + 1) + 2T_{RD} + T_{IP1},$$

which omits the guard interval T_{GD} that separates the CS_SYNC packet from the tone sequence in *each* of the two bursts. With the series’ fixed $T_{GD} = 10 \mu s$, the omission understates every Mode-3 step by $2T_{GD} = 20 \mu s$. For the campaign report’s own configuration this is $254 \mu s$ against a correct $274 \mu s$, an error of 7.9% in step duration that propagates into subevent occupancy, steps per subevent and radio duty cycle.

Confirmed against the specification. Core 6.3 Table 4.10 shows the guard interval explicitly in *both* extended packets [Core 6.3, Vol. 6, Part H, Sec. 4.3.4], and Section 2.6 fixes $T_{GD} = 10 \mu s$ “independent of the LE PHY being used”. This report uses Eq. (1), which includes both. The campaign report should be corrected if its Mode-3 numbers are relied upon for scheduling.

4 Position of Mode-3 in the CS subevent

Figure 1 places Mode-3 in the causal chain. Mode-0 still runs first and still supplies the subevent frequency reference FFO_{RX}^* , which Mode-3 needs for three separate purposes: carrier placement for both the packet and the tone, rescaling of the reflector’s reported turnaround interval in the timing chain, and bounding of the drift term in the phase chain [Core 6.3, Vol. 6, Part H, Sec. 4.3.1; Vol. 6, Part A, Sec. 3.5].

The structural difference from a campaign that interleaves Mode-1 and Mode-2 steps is visible in the figure: the two chains diverge and reconverge *within one channel*. In an interleaved campaign the reconvergence happens only at the procedure level, after aggregation, because the round-trip time was measured on a different channel at a different time.

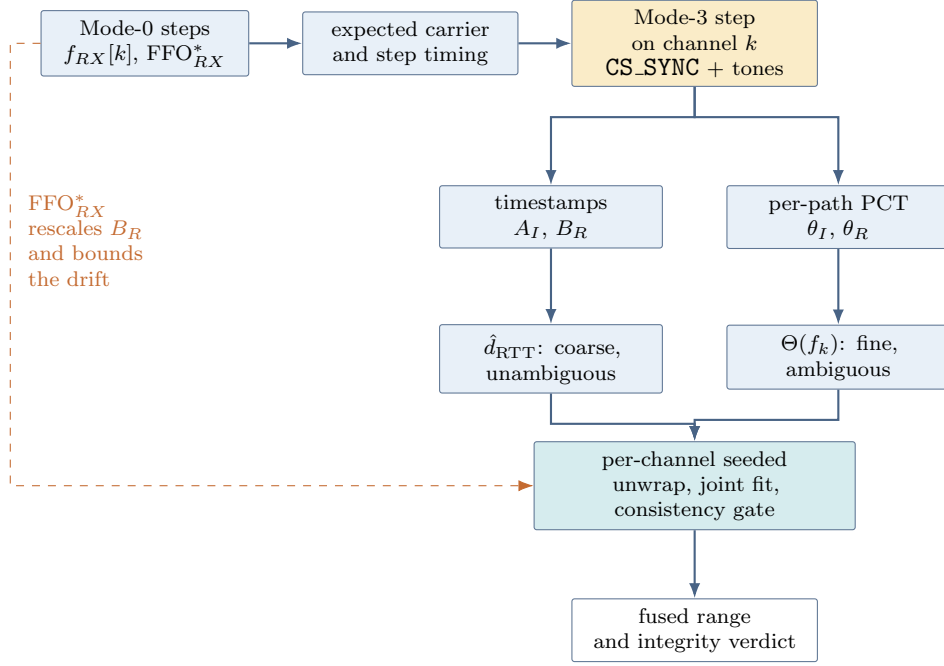


Figure 1: Position of Mode-3. A single step forks into the timing and phase chains, which rejoin at a fusion stage that is now *per channel* rather than per procedure, because both observables were acquired on the same channel in the same step.

5 Mode-3 step structure and timing

5.1 The extended packet and its mirrored order

A Mode-3 step carries an *extended packet* in each direction [Core 6.3, Vol. 6, Part H, Sec. 4.3.4]. The initiator's extended packet is a CS_SYNC packet, then a guard interval, then a tone sequence covering every antenna path. The reflector's extended packet contains the same three elements in the reverse order: tone sequence, guard interval, CS_SYNC. Both are transmitted on the same channel, and each is followed by the transmitter ramp-down period.

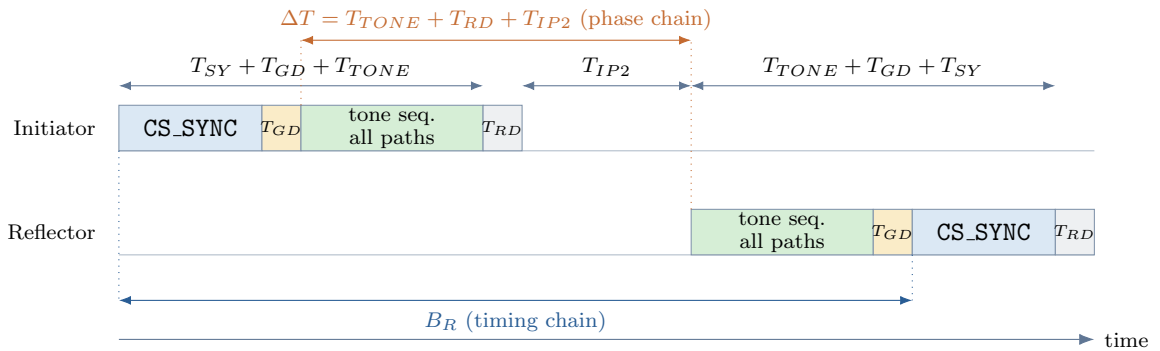


Figure 2: Mode-3 step. The initiator sends CS_SYNC, guard, tone sequence; the reflector sends tone sequence, guard, CS_SYNC. The mirrored order places the two tone sequences *adjacent across the interlude*, so the phase chain's reciprocal interval ΔT is as short as it is in a pure Mode-2 step, while the two CS_SYNC packets are pushed to the outer edges, lengthening the reflector turnaround B_R seen by the timing chain.

5.2 Step duration and the two derived intervals

Writing $T_{TONE} = (T_{SW} + T_{PM})(N_{AP} + 1)$ for the tone-sequence duration [Core 6.3, Vol. 6, Part H, Secs. 4.3.3 and 4.5], the nominal step duration is

$$T_{step,3} = 2[T_{SY} + T_{GD} + T_{TONE} + T_{RD}] + T_{IP2}, \quad (1)$$

with the frequency-change period T_{FCS} inserted between scheduled steps that use different channels [Core 6.3, Vol. 6, Part H, Sec. 4.3]. Two further intervals, neither of which is $T_{step,3}$, govern the two error chains.

The phase chain is governed by the interval between the reflector's measurement of the initiator's tone on antenna path p and the initiator's measurement of the reflector's tone on the same path. The initiator's slot p begins at $T_{SY} + T_{GD} + (p-1)(T_{SW} + T_{PM})$; the reflector's burst begins at $T_{SY} + T_{GD} + T_{TONE} + T_{RD} + T_{IP2}$ and its slot p begins $(p-1)(T_{SW} + T_{PM})$ later. Subtracting,

$$\Delta T = T_{TONE} + T_{RD} + T_{IP2}, \quad (2)$$

independent of p , and of T_{SY} and T_{GD} .

The timing chain is governed by the reflector turnaround, from the arrival of the initiator's CS_SYNC to the departure of its own. Reading Figure 2,

$$B_R = T_{SY} + 2T_{GD} + 2T_{TONE} + T_{RD} + T_{IP2}. \quad (3)$$

Table 3: Mode-3 timing parameters. Permitted sets are configuration-dependent and subject to both devices' declared capabilities; read the value lists from the specification revision in force [Core 6.3, Vol. 6, Part H, Sec. 4.3; Vol. 4, Part E, CS configuration]. The three fixed values are those used consistently across this report series.

Parameter	Meaning	Representative permitted values
T_{SY}	CS_SYNC packet duration	PHY- and RTT-type-dependent; LE 2M: 26, 42, 58, 74, 90 μ s
T_{GD}	Guard interval between CS_SYNC and tone sequence	10 μ s(fixed in this series)
T_{SW}	Antenna path switch period, including RF and phase settling	1, 2, 4, 10 μ s when switching is performed; 0 with a single antenna
T_{PM}	Phase measurement period	10, 20, 40 μ s (40 μ s mandatory)
T_{RD}	Transmitter ramp-down	5 μ s(fixed)
T_{IP2}	Interlude period, confirmed for Mode-3 (§3)	10, 20, 30, 40, 50, 60, 80, 145 μ s (145 μ s mandatory)
T_{FCS}	Frequency-change period between steps	15, 20, 30, 40, 50, 60, 80, 100, 120, 150 μ s

Reading the table against Eqs. (2) and (3) shows how differently the two chains respond to configuration. Lengthening the CS_SYNC packet from 42 to 74 μ s — moving from a 32-bit to a 96-bit sounding sequence — improves the timing chain's thermal precision by $\sqrt{3}$ and leaves ΔT untouched, so it is close to free for the phase chain. Enlarging the tone sequence, by contrast, lengthens *both* ΔT and B_R , and does so twice over in the latter.

Table 4: Step duration, phase reciprocal interval and reflector turnaround from Eqs. (1)–(3), with $T_{GD} = 10 \mu s$, $T_{RD} = 5 \mu s$ and the LE 2M PHY. The Mode-1 and Mode-2 columns use the companion reports’ formulas at the same settings.

T_{SY}	N_{AP}	T_{SW}	T_{PM}	T_{IP2}	$T_{step,3}$	ΔT	B_R
42	4	2	10	40	$274 \mu s$	$105 \mu s$	$227 \mu s$
42	1	2	10	40	$202 \mu s$	$69 \mu s$	$155 \mu s$
42	4	2	10	145	$379 \mu s$	$210 \mu s$	$332 \mu s$
74	4	2	10	40	$338 \mu s$	$105 \mu s$	$259 \mu s$
42	4	10	20	40	$454 \mu s$	$195 \mu s$	$407 \mu s$
42	4	2	10	10	$244 \mu s$	$75 \mu s$	$197 \mu s$

6 Why the sub-slot order is mirrored

The mirroring of the two extended packets is the single design decision that distinguishes Mode-3 from a naive concatenation of a Mode-1 and a Mode-2 step, and it is worth deriving rather than asserting.

6.1 The counterfactual

Suppose both devices transmitted CS_SYNC first and the tone sequence second. Repeating the arithmetic of Section 5 for that hypothetical ordering,

$$\Delta T^{(\text{un-mirr.})} = B_R^{(\text{un-mirr.})} = T_{SY} + T_{GD} + T_{TONE} + T_{RD} + T_{IP2}, \quad (4)$$

so the two intervals would be equal. Mirroring shortens the first by $T_{SY} + T_{GD}$ and lengthens the second by $T_{GD} + T_{TONE}$. For the configuration of Section 12 this moves ΔT from $157 \mu s$ to $105 \mu s$, a reduction of 33%, and B_R from $157 \mu s$ to $227 \mu s$, an increase of 45%.

6.2 Why that is the right trade

Both intervals enter their respective chains through the same physical mechanism — a residual fractional frequency offset ϵ integrated over an interval — and with the same coefficient. From the companion reports, the phase chain acquires a bias $c\epsilon\Delta T/2$ (Mode-2 report, Eq. for δd_ϵ) and the timing chain acquires a bias $c\epsilon B_R/2$ (Mode-1 report, clock-scale correction). The two are numerically identical functions of their intervals. What differs is the *budget each is spent against*.

Table 5: Effect of the mirrored ordering on the residual-frequency bias in each chain, for the worked configuration of Section 12. The final columns express each change as a fraction of that chain’s total uncertainty from Table 11: 15.7 mm for the phase chain and 128 mm for the timing chain.

Residual offset	Phase chain, $c\epsilon\Delta T/2$		Timing chain, $c\epsilon B_R/2$	
	un-mirrored	mirrored	un-mirrored	mirrored
1 ppm	23.53 mm	15.74 mm	23.53 mm	34.03 mm
0.1 ppm	2.35 mm	1.57 mm	2.35 mm	3.40 mm
0.05 ppm	1.18 mm	0.79 mm	1.18 mm	1.70 mm
Change at 1 ppm	−7.79 mm, −49.6% of budget		+10.49 mm, +8.2% of budget	

The mirroring moves error from the chain that cannot afford it to the chain that can. At a Mode-0 residual of 1 ppm, mirroring removes 7.8 mm from a phase chain whose entire remaining budget is 15.7 mm — halving it — and adds 10.5 mm to a timing chain whose budget is 128 mm, which is an eighth of a standard deviation. The absolute amount of error added exceeds the amount removed; that is irrelevant, because the two are not fungible. The phase estimate is the reported range and the timing estimate is only a seed and a sanity check, and a seed needs to be right to within tens of centimetres, not millimetres.

A second consequence follows from Eq. (2): ΔT is independent of T_{SY} and T_{GD} . The packet can therefore be lengthened freely — a longer sounding sequence, a 96-bit or 128-bit RTT type — to improve the timing chain, and the phase chain does not notice. In the un-mirrored ordering, every microsecond added to the packet would have been paid for twice by the phase chain. Mirroring is what makes the packet length a free parameter.

What does *not* change. The reflector’s turnaround does not need to be known a priori in either ordering: the reflector *measures* B_R against its own clock and reports it, so the interval cancels in the reciprocal construction exactly as it does in Mode-1. Only the residual *scale* error on that measured interval survives, which is the ϵB_R term above. A longer B_R is therefore not a timing error in itself; it is only a lever arm for a clock error.

7 What same-step acquisition provides

Everything in this section is a consequence of one fact: in a Mode-3 step the round-trip time and the phase samples traverse the same hardware, on the same channel, in the same antenna path order, within a single step. In a campaign that interleaves Mode-1 and Mode-2 steps, none of this holds.

Table 6: What is shared between the two observables in a Mode-3 step, and what an interleaved Mode-1/Mode-2 campaign must instead assume.

Quantity	Mode-3: same step	Interleaved Mode-1 / Mode-2
Channel	Identical by construction	Different channels; per-channel calibration differences enter
Instant	Within $T_{step,3}$, typically $< 300 \mu s$	Separated by one or more step periods, typically milliseconds
Geometry	Frozen: sub-millimetre change at any plausible velocity	May differ by centimetres at walking speed
Antenna path	Same permutation, same physical path	Independent permutations; paths must be matched by index
Gain state	One AGC decision covers both	Two independent AGC decisions
Local oscillator	One continuous run	Retuned between steps
Temperature	Identical	Drifts across a procedure
Integrity	Packet and tone share a step; see §10	Tone unprotected by any packet in its own step

7.1 Geometry is frozen within the step

The Mode-2 report establishes that coherent assembly of phase samples is valid only while the geometry is stable to a small fraction of a wavelength, and that at 2.44 GHz a relative velocity of 1 m/s covers a tenth of a wavelength in 12 ms. Within one Mode-3 step of $274 \mu\text{s}$ the same velocity moves the range by 0.27 mm, and at a vehicular 30 m/s by 8.2 mm. The round-trip time and the phase sample of a given step therefore describe the same distance to well within the phase chain’s own precision, at any speed at which Channel Sounding is usable.

This is what licenses the per-channel operations of Section 8. In an interleaved campaign the round-trip time used to seed channel k ’s unwrap was measured milliseconds earlier, and at 1 m/s that is already centimetres of geometry change — harmless for seeding, but fatal for any attempt to use the round-trip time as a per-channel constraint in a joint fit.

7.2 One hardware traversal, two calibrations

The Mode-1 report shows that the four transmit and receive chain *group delays* accumulate into an additive loop delay D_{loop} , whose uncalibrated value of 10 ns costs 1.50 m of range. The Mode-2 report shows that the four chain *phase responses* accumulate into $\Phi_{loop}(f)$, whose linear part is the same physical quantity: $\Phi_{loop}(f) \approx \Phi_0 + 2\pi f D_{loop}$.

In a Mode-3 step both are traversed once, on one channel, in one gain state. Three consequences:

1. **The calibrations are not independent and should not be stored independently.** The slope of Φ_{loop} over the measured band and the scalar D_{loop} of the timing chain are two estimates of one quantity. Storing them as one surface with a consistency constraint halves the calibration state and exposes errors in either.
2. **Each calibration can be verified against the other in the field.** A conducted delay-line test measures both at once; a drift in one that is not matched by a drift in the other indicates that something other than propagation delay has changed — most often a gain-state change that one chain saw and the other did not.
3. **The residual is bounded by the tighter of the two.** An implementation that has calibrated $\Phi_{loop}(f)$ to the 0.1 ns residual the phase chain requires has, for free, a D_{loop} far better than the 0.5 ns the timing chain needs.

The guard interval is a calibration boundary, not just dead time. T_{GD} exists because the transmitter must transition between a modulated packet and an unmodulated carrier, and the receiver must follow. If that transition changes any state that affects phase — a filter bandwidth, a gain, a synthesiser mode — then the packet and the tone in the *same* step traverse subtly different paths, and the shared-calibration argument above weakens exactly where it is most valuable. Verifying that the two sub-slots see the same chain is a Mode-3-specific test with no Mode-1 or Mode-2 equivalent; it appears as item 5 in Section 14.

8 Joint estimation

8.1 Per-channel seeded unwrapping

The Mode-2 report identifies sequential phase unwrapping as the fragile step in the phase chain: it fails when the true channel-to-channel phase step exceeds π , which for adjacent 1 MHz channels occurs beyond $c/(4\Delta f) = 74.95 \text{ m}$ and, across a gap of G channels, beyond

74.95/G m. It also fails on any single faded channel, and one failure propagates as a step to every subsequent channel.

With a same-step round-trip time the sequential accumulation is replaced by a set of independent decisions. The range is recovered from the *slope* of Θ against frequency, so what must be resolved is not the absolute fringe number at any channel but the fringe *increment* between consecutive measured channels. Writing $\Delta_k = f_k - f_{k-1}$ for the actual spacing, which may span a gap in the channel list,

$$\Delta\Theta_k = \Theta_{cal}(f_k) - \Theta_{cal}(f_{k-1}), \quad \hat{n}_k = \left\lfloor \frac{1}{2\pi} \left(\frac{4\pi\Delta_k \hat{d}_{RTT}}{c} - \Delta\Theta_k \right) + \frac{1}{2} \right\rfloor, \quad (5)$$

$$\Theta_{unw}(f_k) = \Theta_{unw}(f_{k-1}) + \Delta\Theta_k + 2\pi\hat{n}_k. \quad (6)$$

with $\hat{d}_{RTT}$ the round-trip-time range from the same step. Each increment is decided against the seed rather than against the accumulated history, so a faded channel corrupts only its own two increments instead of every channel that follows. The condition for correctness is not a distance limit but a *seed accuracy* limit,

$$\sigma_{RTT} < \frac{1}{k_\sigma} \cdot \frac{c}{4\Delta_{max}}, \quad (7)$$

where Δ_{max} is the largest spacing present and k_σ the desired confidence in standard deviations.

Mode-3 makes differential unwrapping unconditionally safe. Rearranging Eq. (7), the worked example's 12.8 cm seed supports correct nearest-integer decisions at 3σ for any spacing up to $\Delta_{max} = c/(4k_\sigma\sigma_{RTT}) = 195$ MHz — about two and a half times the entire width of the 2.4 GHz band. Gaps in the channel list, however large, therefore cease to be a source of unwrap error — including the CS raster's own 4 MHz gap between 2424 and 2428 MHz [Core 6.3, Vol. 6, Part A, Sec. 2], which by itself caps sequential unwrapping at 18.74 m in a pure Mode-2 campaign. This is the most operationally valuable property of Mode-3, and an interleaved campaign cannot reproduce it: there the seed comes from a different channel at a different time, so it can bound the aggregate range but cannot be attached to an individual channel's increment.

The absolute fringe remains out of reach, and that is fine. Eq. (5) is deliberately differential. Resolving the *absolute* fringe number at a channel — roughly 162 at 2.44 GHz and 10 m — would require $\sigma_{RTT} < c/(4k_\sigma f_c) = 1.0$ cm at 3σ , which is an order of magnitude beyond what any round-trip-time chain delivers. No slope-based range estimator needs it. An implementation that does need absolute phase — some super-resolution formulations, or any attempt to combine assemblies coherently across time — must obtain it another way, and should not assume the Mode-3 seed provides it.

8.2 The combined aperture

The two sub-slots sample the channel transfer function differently and complementarily. The tone gives one exact frequency-domain sample $H(f_k)$ at the channel centre, as developed in the Mode-2 report. The CS_SYNC packet occupies roughly 2 MHz *around* f_k , and the group-delay regression of the Mode-1 report is precisely an estimate of $\arg H(f)$ across that 2 MHz. A Mode-3 step therefore yields, per channel, a narrow contiguous band of phase information as well as a point sample.

Assembling these across K channels gives a frequency-domain data set that is denser than the tone samples alone. Two uses follow.

- **Sidelobe suppression in the transform domain.** The Mode-2 report notes that gaps in the channel list raise the sidelobes of the inverse transform, and that a strong

reflector can then capture the first-path search through a sidelobe. Packet-derived phase partially fills the gaps between tone samples, and the resulting effective aperture has a better-conditioned point spread function.

- **Extension of the unambiguous span.** The tone samples alone repeat every $c/(2\Delta f) = 149.90$ m. The packet band provides phase information at spacings far finer than Δf , whose own ambiguity is correspondingly much larger. Combining the two is the classical coarse–fine construction, and it delivers the phase chain’s precision over the packet band’s unambiguous span without recourse to the round-trip time at all — though in Mode-3 the round-trip time is available anyway and is simpler to use.

8.3 Joint likelihood

Where computation permits, the principled formulation estimates one delay from both observables at once. With $z_k[n]$ the packet samples on channel k , s the known CS_SYNC reference, $H[k]$ the tone-derived complex sample, and τ the common two-way delay,

$$\hat{\tau} = \arg \max_{\tau} \left[\underbrace{\sum_k \lambda_k \left| \sum_n z_k[n] s^* \left(\frac{n}{F_{s,cal}} - \tau \right) \right|^2}_{\text{packet term, Mode-1 report}} + \underbrace{\mu \left| \sum_k w_k H[k] e^{j2\pi f_k \tau} \right|^2}_{\text{tone term, Mode-2 report}} \right], \quad (8)$$

with λ_k and μ chosen from the respective noise variances so that each term is weighted by its own information content. The packet term is broad and unambiguous; the tone term is sharp and periodic. Their sum is a single objective whose global maximum is the sharp peak nearest the broad one — which is exactly the coarse–fine fusion of Section 8, expressed as one estimator rather than two stages.

Equation (8) is a reference formulation, not a recommendation for every implementation. The two-stage version — estimate the round-trip time, seed the unwrap, fit the slope — reaches essentially the same answer at a small fraction of the cost, and is what most implementations should build. The joint form earns its cost in two situations: when the tone samples are sparse enough that the coarse–fine gap is not comfortably bridged, and as a reference against which the two-stage implementation is validated.

8.4 Consistency as an estimator input, not only a check

Because both chains produce a range for the *same* channel and step, their difference

$$\delta_k = \hat{d}_{\text{RTT},k} - \hat{d}_{\Theta,k} \quad (9)$$

is a per-step observable in its own right, with a known distribution under the null hypothesis of a clean single-path channel: zero mean and variance $\sigma_{\text{RTT}}^2 + \sigma_{\Theta}^2$. Departures are informative rather than merely alarming.

The third row deserves emphasis: a residual that grows linearly with channel frequency is the signature of the ϵ term, and because the two chains weight ϵ by *different* intervals — ΔT for phase, B_R for timing — the ratio of the two chains’ errors is the known constant $B_R/\Delta T$. A Mode-3 implementation can therefore estimate the residual frequency offset from the ranging data itself, as a cross-check on Mode-0, which neither constituent mode can do alone.

Table 7: Interpreting the per-step cross-observable residual of Eq. (9).

Observed pattern of δ_k	Most likely cause	Action
Within $\pm k_\sigma$, unstructured	Clean channel	Accept; use the phase estimate
Constant offset across all channels	Loop-delay or loop-phase calibration error	Recalibrate; the offset is the correction
Offset proportional to f_k	Residual ϵ , or wrong interlude parameter (§3)	Check Mode-0 sign and the T_{IP2} configuration
Large on isolated channels only	Interference or fading on those channels	Gate those channels
Exactly one fringe, $\pm c/(2\Delta f)$	Unwrap slip despite seeding	Re-seed; check TQI on that channel
Phase estimate consistently short	Multipath capture by the slope fit	Switch to transform-domain first path
Large, one-sided, with good TQI	Possible tone injection	Check NADM; see §10

9 Precision of the two chains

Mode-3 does not improve the precision of either observable. Both bounds are those of the companion reports, evaluated at the Mode-3 configuration, and are restated here only so that the two can be compared on one page.

9.1 Timing chain

From the Mode-1 report, for a known signal in additive white Gaussian noise,

$$\sigma_\tau \geq \frac{1}{2\pi \beta_{rms} \sqrt{2N\gamma}}, \quad \sigma_{RTT} = \sqrt{2} \sigma_\tau, \quad \sigma_{d,RTT} = \frac{c}{2} \sigma_{RTT}, \quad (10)$$

with β_{rms} the root-mean-square bandwidth of CS_SYNC, N the number of modulated symbols set by the RTT type, and γ the per-symbol signal-to-noise ratio.

The timing figures here assume the LE 2M PHY, not LE 2M 2BT. Core 6.3 defines a third PHY usable with CS_SYNC: LE 2M 2BT, at 2 Mb/s with a bandwidth-bit-period product of $BT = 2.0$ rather than 0.5, and a minimum frequency deviation at the symbol centre of 420 kHz [Core 6.3, Vol. 6, Part A, Secs. 1 and 3.1.2]. A $BT = 2.0$ Gaussian pulse is far smoother, so its root-mean-square bandwidth — and hence the timing precision of Eq. (10) — differs from the $\beta_{rms} = 0.50$ MHz used throughout this report and the companion Mode-1 report. All timing-chain numbers here are therefore specific to the LE 2M PHY. An implementation using LE 2M 2BT must re-derive β_{rms} from that pulse shape before applying any of them.

Averaging M valid steps reduces $\sigma_{d,RTT}$ by $\sqrt{M}$ but does not touch D_{loop} error or multipath bias.

9.2 Phase chain

From the Mode-2 report, for a tone of energy $\mathcal{E}$ in noise of density N_0 ,

$$\sigma_\theta \geq \frac{1}{\sqrt{2\mathcal{E}/N_0}}, \quad \mathcal{E}/N_0 = \gamma B_n T_{PM}, \quad \sigma_{d,\Theta} = \frac{c}{4\pi} \frac{\sqrt{2} \sigma_\theta}{\sqrt{\sum_k (f_k - \bar{f})^2}}. \quad (11)$$

9.3 The two together

Table 8: Thermal precision of both chains at the worked configuration — LE 2M PHY, 32-bit sounding sequence ($T_{SY} = 42 \mu s$, $N = 32$), $T_{PM} = 10 \mu s$, $\gamma = 20$ dB, all 72 CS channels spanning 74 MHz. Each Mode-3 step contributes one round-trip time *and* one phase sample, so the two columns describe the same 72 steps.

Quantity	Timing chain	Phase chain
Per-step / per-tone precision	$\sigma_\tau = 3.98$ ns	$\sigma_\theta = 1.28^\circ$
Two-way observable precision	$\sigma_{RTT} = 5.63$ ns	$\sigma_\Theta = 1.81^\circ$
Single-step range precision	0.844 m	—
Range precision, all 72 steps	9.94 cm	0.407 cm
Resolution	≈ 75 m	2.03 m
Ambiguity	none	149.90 m
Residual-frequency lever arm	$B_R = 227 \mu s$	$\Delta T = 105 \mu s$

The ratio of the two range precisions is about 23, and it is the reason the architecture is hierarchical: the phase chain reports the range, the timing chain seeds and audits it.

Antenna paths help the phase chain and not the timing chain. With $N_{AP} = 4$ the step yields four phase samples per channel but still only one round-trip time, because the CS_SYNC exchange is not repeated per path. Combining four paths after removing their calibrated offsets could in principle halve $\sigma_{d,\Theta}$ to 0.21 cm; the paths are not statistically independent — they share the oscillator, the step and much of the multipath — so the realised gain is smaller, and the offsets must be known to better than the precision being claimed. Treating the extra paths as diversity, selecting on TQI, is the robust use and is usually worth more than the averaging.

10 Same-step integrity

The Mode-2 report establishes the asymmetry that governs this section. A CS_SYNC packet carries a per-step access address from the CS deterministic random bit generator and, optionally, a random or marker-bearing sequence, so an attacker cannot synthesise a valid early arrival; the Normalized Attack Detector Metric quantifies departures from the expected waveform [Core 6.3, Vol. 6, Part H, Sec. 4.4]. A tone has none of these properties: it is an unmodulated carrier at a known frequency, fully predictable, and an attacker can transmit one at arbitrary phase. Phase ranging alone is not a distance-bounding measurement.

Mode-3 changes what can be done about this, in a way that is worth stating precisely because it is easy to overclaim.

10.1 What Mode-3 does provide

1. **Per-channel binding.** The tone that produced $\Theta(f_k)$ and the packet that produced NADM for that channel are in the same step, microseconds apart, on the same carrier. An attacker who injects a tone to shift the phase on channel k must, in the same step, either leave the packet alone — in which case the cross-observable residual δ_k of Eq. (9) exposes the injection — or also forge the packet, which requires defeating the unpredictable access address and the NADM check.
2. **A quantitative gate.** The residual δ_k has a known null distribution, so the detector has a calculable false-alarm rate rather than a heuristic threshold. At the worked example's

numbers the 3σ gate is 39 cm (Section 12), which bounds the distance reduction an undetected tone-only attack can achieve on any single channel.

3. **No timing gap to exploit.** In an interleaved campaign, an attacker can behave honestly during Mode-1 steps and inject only during Mode-2 steps, because the two are separated in time and the consistency check is only available in aggregate. In Mode-3 there is no step in which the tone is present and the packet is not.

10.2 What it does not provide

Mode-3 does not make the phase observable itself unpredictable, and the gate of item 2 above is a *per-channel* bound, not a bound on the aggregate. An attacker who shifts every channel by less than the gate can still bias the fitted slope. The residual budget should therefore be set from the aggregate consequence — how much range error survives K channels each shifted by up to the per-channel gate — and not from the per-channel figure alone. The channel hopping sequence and the antenna path permutation remain material defences, because an attacker must track both to inject coherently, and neither should be weakened for implementation convenience [Core 6.3, Vol. 6, Part H, Secs. 4.2 and 4.5].

The honest claim. Mode-3 is the configuration in which phase-based ranging can be given a defensible integrity story, because every phase sample is accompanied by an unpredictable waveform measured through the same chain at the same instant. It is not a distance-bounding protocol on its own, and a security-relevant decision should still be gated on the packet’s NADM and on the cross-observable residual together, with the threshold justified by the aggregate analysis above.

11 Air-time economics: when to configure Mode-3

Mode-3’s step is longer than either constituent, so whether it is cheaper than the alternative depends entirely on how often the campaign would otherwise schedule a round-trip-time step. Using the occupancy convention of the companion campaign report, $T_{occ} = T_{step} + T_{FCS}$, and its configuration — $T_{SY} = 42 \mu s$, $N_{AP} = 4$, $T_{SW} = 2 \mu s$, $T_{PM} = 10 \mu s$, interlude $40 \mu s$, $T_{FCS} = 80 \mu s$:

Table 9: Air-time comparison at the campaign report’s configuration. Step durations from Eq. (1) and the companion reports’ formulas; occupancy adds $T_{FCS} = 80 \mu s$.

Scheduling pattern	Step	Occupancy	Observables produced
One Mode-1 step	$134 \mu s$	$214 \mu s$	1 round-trip time
One Mode-2 step	$170 \mu s$	$250 \mu s$	1 phase sample
One Mode-3 step	$274 \mu s$	$354 \mu s$	1 round-trip time + 1 phase sample
Mode-1 + Mode-2 separately	$304 \mu s$	$464 \mu s$	1 of each, different channels
Mode-3	$274 \mu s$	$354 \mu s$	1 of each, <i>same</i> channel
<i>Saving</i>	$30 \mu s$ (9.9%)	$110 \mu s$ (23.7%)	and same-step coherence
4 channels, Mode-2 main with 1 Mode-1 sub	—	$1214 \mu s$	4 phase + 1 round-trip time
4 channels, all Mode-3	—	$1416 \mu s$	4 phase + 4 round-trip times
<i>Cost</i>	—	$+202 \mu s$ (+16.6%)	for $4\times$ the timing data

Two conclusions follow, and they point in opposite directions depending on the campaign pattern.

Against a one-for-one alternative, Mode-3 is strictly better. If a design genuinely needs a round-trip time on every channel, Mode-3 delivers it for 23.7% less occupancy than two separate steps, and delivers it with same-step coherence that the two separate steps cannot provide at any price. The saving comes mostly from paying one frequency-change period instead of two, which is why the occupancy saving (23.7%) is more than twice the step-duration saving (9.9%).

Against a sparse-sub-mode campaign, Mode-3 costs more. The campaign report's own pattern — Mode-2 as the main mode with a Mode-1 sub-mode every fourth step — buys ambiguity resolution and integrity at a low duty. Replacing it with all-Mode-3 costs 16.6% more occupancy. The question is therefore what the additional round-trip times are worth, and the answer is given by Section 8: they convert unwrapping from a procedure-level assumption into a per-channel decision, and they convert the integrity check from an aggregate to a per-channel gate.

A decision rule. Configure Mode-3 when any of the following holds: the channel list has large or unpredictable gaps, so that per-channel unwrap seeding is needed (Section 8); the application is security-relevant, so that per-channel binding of tone to packet is needed (Section 10); the target may move fast enough that a round-trip time from a different step describes a different geometry; or the design would otherwise schedule a round-trip-time step for every phase step anyway. Otherwise a Mode-2 main mode with a sparse Mode-1 sub-mode remains the cheaper configuration, and the companion campaign report's scheduling analysis applies.

12 Worked numerical example at 2 440 MHz

The configuration is that of the companion campaign report, so that the numbers here can be checked against its scheduling analysis: LE 2M PHY, RTT type 0x01 (32-bit sounding sequence, $T_{SY} = 42 \mu s$), $T_{GD} = 10 \mu s$, $N_{AP} = 4$ (antenna configuration index 3, 4 or 7), $T_{SW} = 2 \mu s$, $T_{PM} = 10 \mu s$, $T_{RD} = 5 \mu s$, interlude $40 \mu s$, $T_{FCS} = 80 \mu s$, an assembly of $K = 72$ channels at 1 MHz spacing over $S = 71$ MHz centred near 2 440 MHz, per-tone and per-symbol signal-to-noise ratio $\gamma = 20$ dB, and a true separation of $d = 10.000$ m.

12.1 Timing

$$T_{TONE} = (T_{SW} + T_{PM})(N_{AP} + 1) = 12 \times 5 = 60 \mu s, \quad (12)$$

$$T_{step,3} = 2(42 + 10 + 60 + 5) + 40 = 274 \mu s, \quad (13)$$

$$\Delta T = 60 + 5 + 40 = 105 \mu s, \quad (14)$$

$$B_R = 42 + 20 + 120 + 5 + 40 = 227 \mu s. \quad (15)$$

The campaign report's formula, which omits the two guard intervals, would give $254 \mu s$; see Section 3.

12.2 The quantities being measured

$$\tau = \frac{10.000}{299\,792\,458} = 33.356410 \text{ ns}, \quad \text{RTT} = 2\tau = 66.712819 \text{ ns}, \quad (16)$$

$$\Theta(f_c) \equiv 4.896351 \text{ rad} \pmod{2\pi}, \quad \Delta\Theta = \frac{4\pi d \Delta f}{c} = 24.0166^\circ \text{ per MHz}. \quad (17)$$

12.3 Timing-chain budget

With $\beta_{rms} = 0.50$ MHz, $N = 32$ and $\gamma = 100$, Eq. (10) gives $\sigma_\tau = 3.979$ ns, $\sigma_{RTT} = 5.627$ ns and a single-step $\sigma_d = 0.844$ m, falling to 9.94 cm over 72 steps.

Table 10: Timing-chain uncertainty. Rows follow the Mode-1 report’s budget, evaluated at the Mode-3 B_R of 227 μs .

Contribution	Time	Range	Character
Thermal noise, 72 steps	0.663 ns	9.94 cm	random
Loop-delay calibration residual	0.500 ns	7.49 cm	bias
Reporting quantisation, two fields	0.204 ns	3.06 cm	random
Clock scale after Mode-0, 0.05 ppm at B_R	0.011 ns	0.17 cm	bias
FAE table quantisation at B_R	0.004 ns	0.05 cm	bias, per channel
Root-sum-square	—	12.82 cm	

12.4 Phase-chain budget

With $\mathcal{E}/N_0 = 100 \times 10^6 \times 10^{-5} = 1000$, Eq. (11) gives $\sigma_\theta = 1.281^\circ$, $\sigma_\Theta = 1.812^\circ$ and $\sigma_{d,\Theta} = 4.28$ mm over 72 channels.

Table 11: Phase-chain uncertainty. Rows follow the Mode-2 report’s budget, evaluated at the Mode-3 ΔT of 105 μs .

Contribution	Phase	Range	Character
Loop group-delay residual (0.1 ns)	—	1.499 cm	bias
Thermal noise, 72 channels	1.81° per tone	0.407 cm	random
Oscillator phase noise (0.5° rms)	0.71° per tone	0.159 cm	partly correlated
Mode-0 residual, 0.05 ppm at ΔT	—	0.079 cm	bias
FAE table quantisation, 0.0156 ppm	—	0.025 cm	bias, per channel
PCT quantisation, 12-bit I/Q	0.008°	0.003 cm	random
Root-sum-square (excluding multipath)	—	1.563 cm	
Multipath, indoor, first-path estimator	—	typ. +5 to 20 cm	one-sided

12.5 The cross-observable gate

The two chains are independent to first order — they share D_{loop} and ϵ , but those enter as biases that the gate is intended to detect rather than as common noise — so the residual of Eq. (9) has

$$\sigma_\delta = \sqrt{\sigma_{d,RTT}^2 + \sigma_{d,\Theta}^2} = \sqrt{12.82^2 + 1.57^2} = 12.91 \text{ cm}, \quad (18)$$

giving a 3σ gate of 38.7 cm and a 4σ gate of 51.7 cm. A step whose two range estimates disagree by more than the chosen gate is a detected fault, not a more precise answer.

Table 12: Deterministic values for the 2 440 MHz example.

Quantity	Value	Unit
$T_{SY} / T_{GD} / T_{SW} / T_{PM} / T_{RD} /$ interlude	42 / 10 / 2 / 10 / 5 / 40	μs
N_{AP}	4	
Tone-sequence duration T_{TONE}	60	μs
Step duration $T_{step,3}$	274	μs
Step duration by the campaign report's formula	254	μs
Occupancy $T_{step,3} + T_{FCS}$	354	μs
Phase reciprocal interval ΔT	105	μs
Reflector turnaround B_R	227	μs
True distance	10.000	m
One-way propagation delay	33.356410	ns
Round-trip time	66.712819	ns
Two-way phase at f_c , modulo 2π	4.896351	rad
Two-way phase step per 1 MHz	24.0166	deg
Timing chain σ_τ	3.979	ns
Timing chain σ_d , single step	0.844	m
Timing chain σ_d , 72 steps	9.94	cm
Timing chain total (RSS)	12.82	cm
Phase chain σ_θ	1.281	deg
Phase chain σ_d , 72 channels	4.065	mm
Phase chain total (RSS)	1.563	cm
Cross-observable σ_δ	12.91	cm
3σ consistency gate	38.7	cm
Max spacing for safe seeded unwrap, 3σ	195	MHz
Seed accuracy needed for the absolute fringe, 3σ	1.02	cm
Bias from 10 ns uncalibrated D_{loop}	1.499	m
Bias from 12 ppm uncorrected offset, timing chain	0.408	m
Bias from 12 ppm uncorrected offset, phase chain	0.189	m

13 Error sources specific to the combined step

The error budgets of the two chains are those of the companion reports and are not repeated. The table below lists only the mechanisms that are new in Mode-3, or whose character changes because the two observables share a step.

The last row is worth expanding. The cross-observable gate of Section 12 assumes the two chains fail independently. They do not fail independently with respect to D_{loop} and ϵ , both of which bias both chains in the same direction. An uncalibrated loop delay shifts both range estimates by $cD_{loop}/2$ and therefore leaves δ_k near zero — the gate is blind to exactly the largest error either chain has. Calibration is a prerequisite for the gate to mean anything, not an alternative to it.

Table 13: Error mechanisms introduced or altered by the combined step.

Source	Effect	Mitigation	Diagnostic
Guard-interval state change	Packet and tone traverse different chains; shared calibration invalid	Verify identical filter, gain and synthesiser state across T_{GD}	Loop delay from packet vs slope of Φ_{loop}
Longer B_R	Clock-scale bias on the timing chain $2.6\times$ Mode-1's	Mode-0 accuracy; shorter interlude	Range versus interlude sweep
Wrong interlude parameter	Step mis-timed; both chains biased	Resolve T_{IP2} per §3	Residual linear in f_k
Missing guard in scheduling model	Subevent overruns; steps dropped	Use Eq. (1)	Actual versus planned steps per subevent
Sub-slot ordering error	Reflector transmits packet first	Conformance check against Fig. 2	ΔT measured $\approx 157\mu s$ not $105\mu s$
Antenna permutation mismatch between sub-slots	Tones attributed to wrong path; packet unaffected	One permutation per step, applied to both	Per-path range consistency
AGC decision split across sub-slots	Packet and tone in different gain states	Freeze gain for the whole extended packet	Range versus input level
LO retune at the guard	Loss of phase reference between packet and tone	Continuous LO across the extended packet	Inter-sub-slot phase repeatability
Seed used for absolute fringe	Gross range error, multiples of $\lambda/2$	Differential unwrap only, Eq. (5)	Range jumps of 6.1 cm
Over-trusted consistency gate	Aggregate attack passes per-channel test	Set gate from aggregate analysis	Slope shift versus per-channel residuals
Correlated biases across chains	D_{loop} and ϵ common to both; gate desensitised	Calibrate first, gate second	Gate residual versus temperature

14 Verification and conformance-oriented test plan

14.1 Layered verification

Items 1–4 of the Mode-1 plan and items 1–4 of the Mode-2 plan apply unchanged to the respective sub-slots and are not repeated. The following are specific to Mode-3.

1. **Structural conformance.** Capture a Mode-3 step off air and verify the sub-slot order in both directions: initiator CS.SYNC, guard, tones; reflector tones, guard, CS.SYNC. Measure $T_{step,3}$, ΔT and B_R against Eqs. (1)–(3). A measured ΔT of $T_{SY} + T_{GD}$ more than predicted is an un-mirrored implementation.
2. **Interlude identification.** Sweep the configured T_{IP1} and T_{IP2} *independently* and observe which one changes the Mode-3 step length. This settles Section 3 empirically for the device under test and should be recorded in the acceptance file.
3. **Guard-interval chain identity.** With a conducted delay line of known length, extract

D_{loop} from the packet sub-slot and the slope of $\Phi_{loop}(f)$ from the tone sub-slot in the same step. They must agree to within the documented residual. Repeat across gain state, channel and temperature; a divergence localises a state change at the guard.

4. **Joint calibration regression.** Verify that the single stored calibration surface reproduces both chains' corrections, and that perturbing it degrades both consistently.
5. **Seeded-unwrap stress.** Construct channel sets with progressively larger gaps, up to and beyond the Δ_{max} of Eq. (7). Verify that the seeded differential unwrap remains correct where predicted and that its failure, when forced, is detected by the consistency gate rather than reported as a range.
6. **Absolute-fringe guard.** Verify that the implementation does *not* attempt to resolve the absolute fringe from the round-trip-time seed. Inject a 5 cm seed error and confirm the reported range moves by 5 cm and not by a multiple of $\lambda/2 = 6.1$ cm.
7. **Cross-observable gate characterisation.** With a clean conducted channel, accumulate the distribution of δ_k over many steps and verify its standard deviation matches σ_δ from the budget. Set the gate from the measured distribution, not the predicted one.
8. **Gate blindness to common bias.** Inject a known D_{loop} error and confirm that δ_k does *not* move — demonstrating, and documenting, that the gate does not detect common-mode calibration error.
9. **Residual-frequency cross-estimate.** Inject a known ϵ of each sign and verify that the ratio of the two chains' induced errors matches $B_R/\Delta T$, and that the implementation's cross-estimate of ϵ recovers the injected value.
10. **Motion within the step.** Sweep relative velocity and confirm that the two chains continue to describe the same geometry, i.e. that δ_k does not develop a velocity-dependent mean, up to the highest supported speed.
11. **Same-step attack injection.** Inject a tone at controlled phase offsets while leaving the packet untouched, and verify that the consistency gate fires at the designed threshold and that NADM remains clean — demonstrating that the two detectors are complementary. Then forge both and verify NADM fires.
12. **Scheduling regression.** Confirm that the controller's own subevent planning uses the step duration of Eq. (1); count actual steps per subevent against the plan.

14.2 Minimum acceptance records

Retain everything the Mode-1 and Mode-2 plans require for the respective sub-slots, and in addition: the measured $T_{step,3}$, ΔT and B_R ; which interlude parameter the device uses; the per-step pairing of round-trip time and phase samples; the joint calibration surface version and both chains' residuals against it; the per-step δ_k series with its measured distribution and the gate in force; the seeded-unwrap decisions with their seeds and margins; and the demonstrated blindness of the gate to common-mode bias.

Table 14: Boundary between specified behaviour and implementation design, for the Mode-3-specific material. The constituent chains’ boundaries are given in the companion reports.

Bluetooth-defined behaviour	Implementation-specific design
Mode-3 step structure: the extended packet and its mirrored sub-slot order [Core 6.3, Vol. 6, Part H, Sec. 4.3.4]	Receive window placement across both sub-slots; capture and buffering strategy
Guard interval T_{GD} between packet and tone	What the radio does during it; whether any state changes
T_{SY} , T_{SW} , T_{PM} , T_{RD} , interlude and T_{FCS} value sets	Which values are chosen, and the resulting $\Delta T/B_R$ trade
Antenna configuration and per-step permutation [Core 6.3, Vol. 6, Part H, Sec. 4.5]	Per-path calibration; diversity, averaging or angle processing
Reported timing differences, PCT, TQI, NADM [Core 6.3, Vol. 4, Part E]	All estimators; quality metrics and thresholds
That both observables arise from one step	Everything built on that fact: seeded unwrap, joint fit, joint calibration, consistency gate
Use of the Mode-0 result for frequency compensation	Rescaling arithmetic; cross-estimation of ϵ from the two chains
Channel selection and hopping [Core 6.3, Vol. 6, Part H, Sec. 4.2]	Assembly into a coherent aperture; gap handling
Architectural role of phase and amplitude information [Core 6.3, Vol. 1, Part A, Sec. 9.2]	The entire ranging algorithm and its fusion policy

15 Normative behaviour versus implementation choice

Four conclusions that prevent common design errors. (1) The sub-slot order is mirrored, and an implementation that concatenates a Mode-1 and a Mode-2 step in the same order in both directions has a ΔT that is $T_{SY} + T_{GD}$ too long — half again as much residual-frequency error in the chain that can least afford it. (2) The step duration includes *two* guard intervals; omitting them understates every step by $2T_{GD}$ and overruns the subevent. (3) The round-trip-time seed supports *differential* unwrapping across arbitrarily large gaps but cannot resolve the absolute fringe, which would need centimetre-accurate timing. (4) The cross-observable consistency gate is blind to the biases that are common to both chains — loop delay and residual frequency — so calibration must precede gating, and the gate must never be presented as covering them.

16 Conclusions

Mode-3 does not measure anything that Mode-1 and Mode-2 cannot measure between them, and it does not measure either quantity more precisely. Its contribution is entirely structural: it places an unambiguous timing observable and a precise phase observable in the same step, on the same channel, through the same hardware, at the same instant, and in doing so converts several procedure-level assumptions into per-channel facts.

The step’s design repays close reading. The mirrored ordering of the two extended packets is not a symmetry for its own sake: it holds the phase chain’s reciprocal interval at the pure Mode-2 value while pushing the timing chain’s reflector turnaround outward, deliberately moving residual-frequency error from the observable that reports the range to

the observable that merely seeds it. It also makes the CS.SYNC length a free parameter for the phase chain, so the timing chain can be strengthened at no phase cost.

What follows from same-step acquisition is the practical payoff. Unwrapping becomes a set of independent nearest-integer decisions that remain correct across any gap the 2.4 GHz band can present, rather than a sequential accumulation that fails on a single faded channel. The loop group delay and the loop phase response become two views of one calibrated hardware path, each able to check the other. And the fully predictable tone acquires, for the first time, an unpredictable companion in its own step, which is what makes a defensible integrity argument for phase-based ranging possible at all.

The costs are honest and quantified. A Mode-3 step is longer than either constituent, the reflector turnaround is two and a half times Mode-1's and correspondingly more demanding of Mode-0, and against a campaign that already uses a sparse round-trip-time sub-mode the extra occupancy is real. Mode-3 is the right configuration when gaps, motion, or integrity make per-channel binding worth paying for, and the companion campaign report's cheaper pattern remains right when they do not.

A Reported quantities at the host interface

A Mode-3 step reports the union of the Mode-1 and Mode-2 field sets for that step [Core 6.3, Vol. 4, Part E, LE CS Subevent Result]. Exact names, widths and encodings are defined in the specification; the summary below makes the data flow concrete.

Table 15: Per-step Mode-3 reported fields.

Field	Encoding	Role
Initiator timing difference	signed, 0.5 ns units	Timing chain: the A_I contribution
Reflector timing difference	signed, 0.5 ns units	Timing chain: the B_R contribution
Packet quality	implementation index	Timing gating
Packet RSSI	dBm	SNR proxy
Packet NADM	ordinal	Integrity gating
Antenna permutation index	index	Maps tone slot to physical path
PCT, per antenna path	12-bit signed I and Q	Phase chain: measured complex tone
TQI, per antenna path	ordinal	Phase gating and weighting
Tone extension indication	flag / count	Noise reference availability
Unavailable indicators	reserved values	Per-observable gating

Three implementation notes are specific to the combined step. First, the two observables have *independent* unavailability indicators: a step whose packet failed may still carry usable tones, and vice versa. An implementation must gate the two chains separately rather than discarding the whole step, or it will throw away good phase data whenever a packet is lost — which, since the phase chain reports the range, is expensive. Second, the pairing of the timing fields with the PCT fields of the same step is what the whole of Section 8 rests on; it must survive the host interface, the transport and any reordering, and it should be verified rather than assumed. Third, the reserved “unavailable” values must be tested for explicitly in both field groups; the Mode-1 and Mode-2 reports each note that interpreting one as a numeric value produces a grossly wrong result, and in Mode-3 such a value would

additionally corrupt the consistency gate and could mask a real fault.

B Numerical test vectors

Vectors B1–B6 of the Mode-1 report and B1–B10 of the Mode-2 report apply to the respective sub-slots. The following exercise the combination.

- C1. Timing algebra.** For $T_{SY} = 42$, $T_{GD} = 10$, $T_{SW} = 2$, $T_{PM} = 10$, $N_{AP} = 4$, $T_{RD} = 5$, interlude 40 (all μs), verify $T_{TONE} = 60$, $T_{step,3} = 274$, $\Delta T = 105$, $B_R = 227$. Repeat for the five further rows of Table 4.
- C2. Mirroring counterfactual.** Recompute with both bursts in packet-first order and verify $\Delta T = 157 \mu s$ and $B_R = 157 \mu s$, and that the phase-chain bias at 1 ppm rises from 15.74 to 23.53 mm.
- C3. Combined single-path channel.** Construct a Mode-3 data set for $d = 10.000$ m over 72 channels: per step a two-way delay of 66.713 ns and a two-way phase step of 24.0166° per MHz. Verify that the timing chain returns 10.00 m to within its noise and the phase chain to within 4.3 mm.
- C4. Seeded unwrap across a gap.** Remove channels to create a single gap of 10, 50, 150 and 250 MHz-equivalent spacing. Verify correct differential unwrapping for the first three and failure only at the fourth, consistent with $\Delta_{max} = 195$ MHz.
- C5. Absolute-fringe guard.** Perturb the seed by 5 cm; the reported range must move by 5 cm, never by 6.1 cm or a multiple thereof.
- C6. Consistency-gate distribution.** With both chains at their budgeted uncertainties, accumulate δ_k over 10^4 steps and verify $\sigma_\delta = 12.91$ cm and a 3σ gate of 38.7 cm.
- C7. Gate blindness.** Inject $D_{loop} = 10$ ns. Both chains must shift by 1.499 m and δ_k must remain within its null distribution, demonstrating the common-mode blindness of Section 14, item 8.
- C8. Cross-estimate of ϵ .** Inject $\epsilon = +1$ ppm. The timing chain must shift by 34.0 mm and the phase chain by 15.7 mm, a ratio of $B_R/\Delta T = 2.162$. Repeat with $\epsilon = -1$ ppm.
- C9. Independent unavailability.** Mark the packet unavailable on a subset of steps and the tones unavailable on a disjoint subset; verify both chains use every step for which their own data is valid, and that the gate is skipped, not failed, where only one is present.
- C10. Air-time regression.** Verify the scheduler’s occupancy for an all-Mode-3 procedure against $354 \mu s$ per step, and against the $1416 \mu s$ per four channels of Table 9.

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